Reversible and Modulated Instance Normalization

Problem formulation and proposed normalization family

Thesis project

Standalone method note

Abstract

This note isolates the scientific formulation from implementation and cluster operations. It is intended to be readable on its own and reusable when drafting a paper introduction or method section.

1 Forecasting task

For an input window $X \in \mathbb{R}^{C\times L}$ and target $Y \in \mathbb{R}^{C\times H}$, a backbone f_θ predicts the future. The study isolates transformations applied around the same backbone and optimization budget.

2 Reversible instance normalization

For each sample and channel, compute location $\mu(X)$ and positive scale $\sigma(X)$. RevIN transforms

$$Z = \gamma \odot \frac{X - \mu(X)}{\sigma(X)} + \nu, \qquad \widetilde{Y} = f_\theta(Z),$$

then returns

$$\widehat{Y} = \sigma(X) \odot \frac{\widetilde{Y} - \nu}{\gamma} + \mu(X).$$

Location may be mean, last value, median, or zero; scale may be standard deviation, median absolute deviation, or one. Affine parameters are optional.

3 Modulated instance normalization

MIN retains the RevIN input transform but learns an independent global output mapping:

$$\widehat{Y} = \sigma(X) \odot \left[\alpha \odot \frac{f_\theta(Z) - \nu}{\gamma} + \beta\right] + \mu(X).$$

The per-channel parameters $(\gamma, \nu, \alpha, \beta)$ start at the identity. Unlike a personalized method, they do not depend on user or cluster.

4 Experimental decomposition

The core family separates no normalization, global standardization, non-affine instance normalization, and affine RevIN. Separate fronts isolate normalized loss, centering/scaling components, robust and nonlinear transformations, and MIN. DLinear and PatchTST remain unchanged external backbones.

5 Evidence required

A valid comparison fixes target splits, query rows, optimizer steps, batch size, initialization distribution, and seeds. Reports distinguish aggregate, equal-user, unseen-user, and worst-user behavior so that a global gain cannot hide population regressions.